

ASSESSMENT TOOL 2 : Scenario-Based

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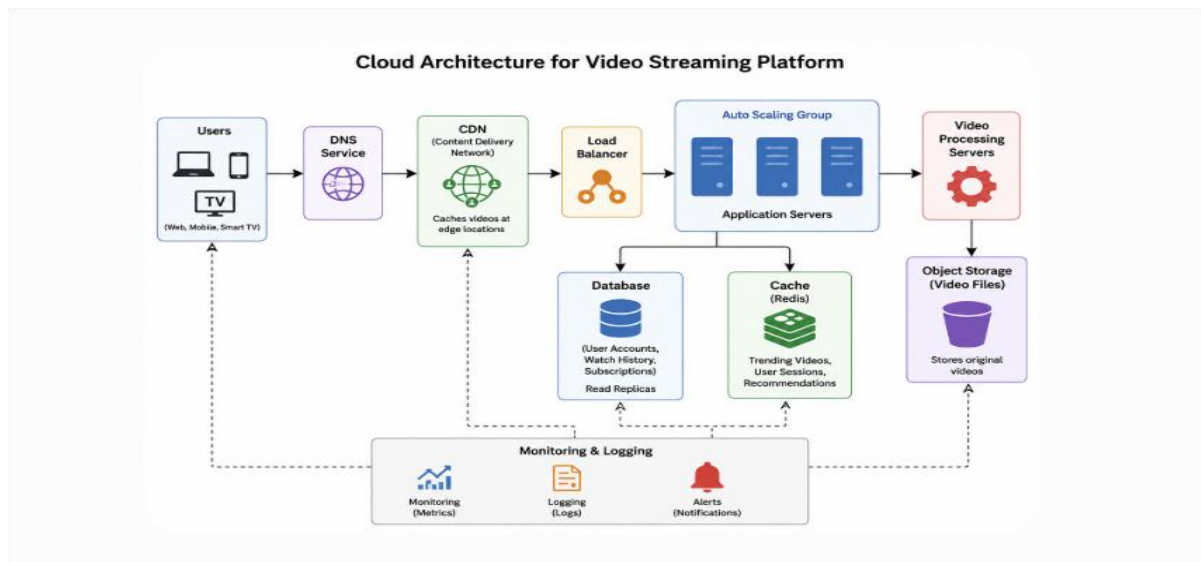
COURSE CODE: CSA15

1.A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Problem Statement

A media company plans to migrate its video streaming platform to the cloud. The platform should support millions of concurrent users, provide minimal buffering, maintain high availability, and optimize infrastructure costs.

Architecture Diagram



Components of the Architecture

1. Users

Users access the streaming platform through mobile apps, smart TVs, or web browsers.

2. DNS Service

Directs users to the nearest cloud region for lower latency.

3. Content Delivery Network (CDN)

- Stores frequently accessed videos at edge locations.
- Delivers videos from the nearest server.
- Reduces buffering and network congestion.

Example: Netflix and YouTube use CDNs worldwide

Load Balancer

The load balancer distributes incoming requests across multiple application servers.

Functions:

- Prevents server overload.
- Ensures equal traffic distribution.
- Removes unhealthy servers automatically

Auto Scaling Group

Application servers automatically increase or decrease based on demand.

For example:

- Normal traffic → 10 servers
- Festival release → 200 servers
- Late night → 20 servers

This provides high performance while reducing unnecessary costs.

Cache (Redis/Memcached)

Frequently accessed data such as:

- User profiles
- Trending videos
- Recommendations

is stored in memory for faster access.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Scenario Understanding

A fintech startup is deploying its applications using Docker containers across multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud Platform (GCP). The company faces challenges in maintaining consistent deployments, managing containerized applications, ensuring high availability, and avoiding vendor lock-in. The solution should provide reliable deployment, scalability, security, and efficient management across multiple cloud environments.

Identification of Risks / Issues

The major issues faced by the fintech startup are:

- Maintaining consistent application deployment across different cloud providers.
- Managing multiple containerized applications efficiently.
- Ensuring high availability during cloud failures.
- Scaling applications automatically during high traffic.
- Service discovery between microservices.
- Data synchronization across clouds.
- Increased operational complexity.

Application of Cloud Concepts

Proposed Solution

The startup should use the following technologies:

- Docker for containerization.
- Kubernetes (K8s) for container orchestration.
- Deploy Kubernetes clusters on AWS, Azure, and Google Cloud.
- Use a Global Load Balancer to route user traffic.
- Use CI/CD pipelines for automated deployments.

Scaling

- Horizontal Pod Auto scaler (HPA) automatically adds or removes pods based on CPU or memory usage.
- Cluster Autoscaler adds or removes worker nodes depending on workload.

This ensures the application can handle increasing user traffic efficiently.

Benefits

- High availability
- Automatic scaling
- Vendor independence
- Disaster recovery
- Better performance
- Easy deployment
- Fault tolerance

Risks

- Higher operational complexity
- Increased networking cost
- Data synchronization challenges

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Scenario Understanding

An e-learning platform has suffered data loss due to accidental deletion of files stored in the cloud. Since students and teachers depend on the platform for learning materials, assignments, and records, the organization needs a cloud-native backup and disaster recovery (DR) solution. The solution should protect data using automated backups, replication, and versioning, minimize downtime, reduce data loss, and ensure continuous availability during failures.

2. Identification of Risks / Issues (20%)

The major risks and issues are:

- Accidental deletion of important data.
- Permanent loss of student records and course materials.
- Service interruption during failures.
- Hardware or cloud region failures.
- Database corruption.

- Delayed recovery affecting online classes.
- Lack of backup and disaster recovery planning.

Application of Cloud Concepts

Proposed Solution

The organization should implement the following cloud-native services:

- Automated Cloud Backup for databases and files.
- Cross-Region Replication to maintain a copy of data in another cloud region.
- Object Storage Versioning to recover deleted or modified files.
- Disaster Recovery Site with automatic failover.
- Cloud Monitoring to detect failures and generate alerts
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Recovery Time Objective (RTO)

Definition: Maximum acceptable time to restore the service after a failure.

- **Achieved By:**
 - Automatic failover.
 - Standby disaster recovery servers.
 - Fast cloud backup restoration.
 - **Example:** RTO = 15 minutes
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Recovery Point Objective (RPO)

- **Definition:** Maximum acceptable amount of data loss.
- **Achieved By:**
- Continuous replication.
- Frequent automated backups.
- Object versioning.
- **Example:** RPO = 5 minutes

Steps to Ensure Business Continuity

- Enable automated cloud backups.
- Enable object storage versioning.
- Configure cross-region replication.
- Implement automatic failover.
- Monitor infrastructure using cloud monitoring tools.
- Perform regular disaster recovery testing.
- Encrypt backup data and control access using IAM.
- Maintain a documented disaster recovery plan.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Scenario Understanding

The government agency needs a hybrid cloud architecture where sensitive government records remain in the on-premises data center, while public websites and non-sensitive applications are deployed in the public cloud. The architecture must provide secure communication, centralized identity management, strong encryption, and compliance with government security policies.

Identification of Risks / Issues (20%)

The major risks and issues are:

- Unauthorized access to sensitive government data.
- Data breaches during communication.
- Compliance and regulatory requirements.
- Secure communication between on-premises and cloud.
- Identity and access management challenges.
- Data synchronization between environment

Application of Cloud Concepts

Hybrid Cloud Architecture

The organization should implement:

- On-Premises Data Center for sensitive applications and databases.
- Public Cloud for public-facing websites and non-sensitive applications.
- VPN or Dedicated Private Connection for secure communication.
- Identity and Access Management (IAM) for centralized authentication.
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Identity Management

A centralized Identity and Access Management (IAM) system should be used.

Implementation

- Centralized user authentication.
- Single Sign-On (SSO) for accessing applications.
- Multi-Factor Authentication (MFA) for enhanced security.
- Role-Based Access Control (RBAC) to grant permissions based on user roles.
- Regular auditing of user activities.

Benefits

- Improved security.
- Easier user management.
- Compliance with government regulations.

Encryption

Data at Rest

- Encrypt databases and storage using **AES-256**.
- Protect encryption keys using a Key Management Service (KMS).

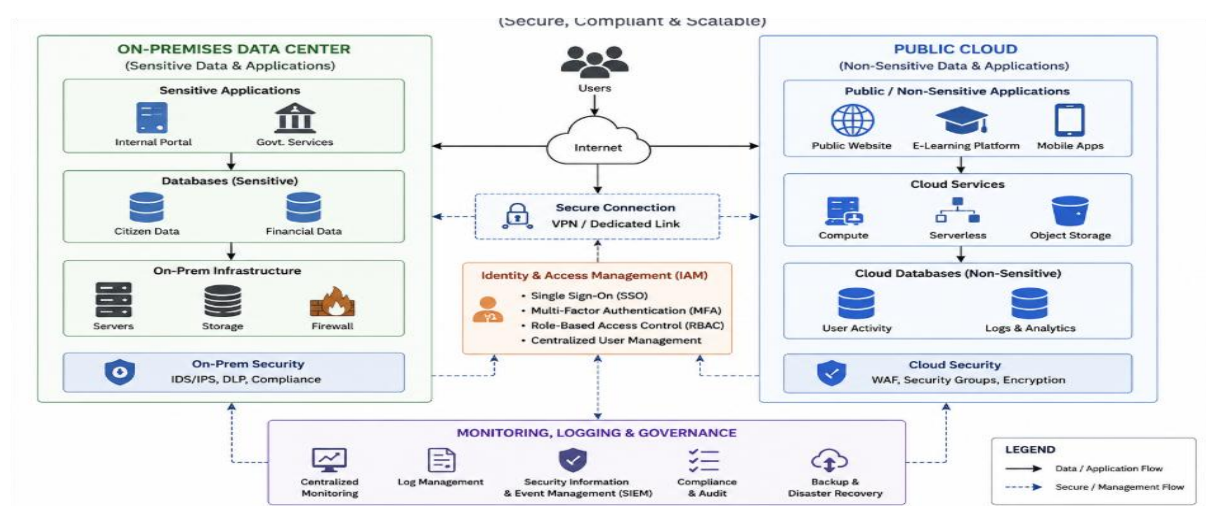
Data in Transit

- Use TLS/SSL encryption.
- Encrypt communication between on-premises and cloud through VPN or dedicated links.

Benefits

- Prevents unauthorized access.
- Protects confidential government information.
- Meets compliance requirements.

Architecture



Challenge

Infrastructure

Complexity

Security Management

Compliance

Network Latency

Explanation

Managing both on-premises and cloud resources is more complex.

Applying consistent security policies across environments.

Meeting government regulations and audit requirements.

Communication between cloud and on-premises may introduce delays.

Challenge	Explanation
Data Synchronization	Keeping data consistent across both environments.
Cost	Additional expenses for networking, monitoring, and management.
Monitoring	Centralized monitoring is required for both environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario Understanding

The SaaS provider serves users from different regions and cannot afford downtime. If one cloud region fails due to a disaster or outage, the application should automatically switch to another region without affecting users. Therefore, the architecture should provide high availability, fault tolerance, disaster recovery, and continuous service.

Identification of Risks / Issues (20%)

The major risks and issues are:

- Regional cloud outage.
- Application downtime.
- Data loss during failures.
- High response time if traffic is not routed properly.
- Database synchronization issues.
- Failure of servers or network components.
- Lack of monitoring and alerting.

Data Replication

- Data should be copied continuously from the primary region to the secondary region.
- Configuration
- Enable cross-region database replication.
- Replicate application files and object storage.
- Use synchronous replication for critical data (if low latency allows).
- Use asynchronous replication for large-scale deployments where performance is a priority.

Benefits

- Prevents data loss.
- Keeps backup region up to date.
- Supports disaster recovery.
- Monitoring and Alerting Strategies

- To detect issues before they affect users, continuous monitoring should be implemented.

Monitoring

- Monitor:
- CPU utilization.
- Memory usage.
- Disk space.
- Network traffic.
- Database performance.
- Application response time.
- Region health.

Alerting

- Configure alerts for:
- Server failures.
- High CPU or memory usage.
- Database replication failures.
- Increased response time.
- Network connectivity issues.
- Storage capacity limits.

Failover Mechanism

When the primary region fails:

1. Health checks detect the failure.
2. Global Load Balancer stops sending traffic to the failed region.
3. Traffic is automatically redirected to the secondary region.
4. Users continue using the application with minimal interruption.

